Representation Learning and Information Extraction in EHR

Laura Igual

UNIVERSITAT DE BARCELONA

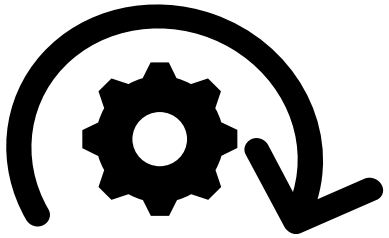
Outline

- Representation Learning in EHR
- Clinical Coding and Medical Concept Representation
- Patient Representation Learning
- Information Extraction in EHR
- More information

Representation Learning in EHR

Representation Learning in EHR

- EHR Data: Why is it challenging?
  - Heterogeneous:
    - Structured: diagnoses, labs, medications
    - Unstructured: clinical notes, reports
    - Temporal: patient history over time
  - High-dimensional & sparse
  - Irregular and incomplete
- Goal: Learn representations that enable:
  - Prediction (mortality, readmission)
  - Patient stratification
  - Clinical decision support



Clinical Coding

Clinical Coding

- An EHR for a particular patient will have patient details (diagnosis, medications,…) as a list of codes.
- There are different **classification schema** (or terminology standards) for:
  - **Diagnoses**:
    International Classification of Diseases (ICD-10)
  - **Procedures**:
    Current Procedural Terminology (CPT)
  - **Laboratory tests**:
    Logical Observation Identifiers Names and Codes (LOINC)
  - **Medications/Drugs**:
    Prescription drugs Normalized (RxNorm, with RxCUIs)
    National Drug Code (NDC)
    Generic Sequence Number (GSN)

CLASSIFICATION SCHEMA EXAMPLES

Schema	Number of Codes	Examples
ICD-10 *(Diagnosis)*	68,000	- J9600: Acute respiratory failure - I509: Heart failure - I5020: Systolic heart failure
CPT *(Procedures)*	9,641	- 72146: MRI Thoracic Spine - 67810: Eyelid skin biopsy - 19301: Partial mastectomy
LOINC *(Laboratory)*	80,868	- 4024-6: Salicylate, Serum - 56478-1: Ethanol, Blood - 3414-0: Buprenorphine Screen
RxNorm *(Medications)*	116,075	- 161: Acetaminophen - 7052: Morphine - 1819: Buprenorphine

Figure source: Schicle *et al*. arXiv:1706.03446v2

International Classification of Diseases (ICD)

- Released by the World Health Organization (WHO) (2019), and updated periodically.
- From ICD-9 (with 14.000 codes) to ICD-10 (with almost 70.000 codes) the specificity has increase and has led to some **unlikely codes**.
- Current version (January 2026): ICD-11.

https://blog.curemd.com/the-most-bizarre-icd-10-codes-infographic/

Other systems

- There are other systems.

  Unified Medical Language System (UMLS) Metathesaurus gathers many health and biomedical standards to enable interoperability between information systems.

- In Spain, codes for diseases and procedures are in the:
  - CLASIFICACIÓN INTERNACIONAL DE ENFERMEDADES 9ª REVISIÓN, MODIFICACIÓN CLÍNICA (CIE-9-MC, which is the **official translation of ICD-9-CM** (an adaptation of the ICD-9)). https://eciemaps.mscbs.gob.es/ecieMaps/browser/index_9_mc.html

  - CLASIFICACIÓN INTERNACIONAL DE ENFERMEDADES 10ª REVISIÓN (CIE-10-ES, which is the **official translation of ICD-10**). https://eciemaps.mscbs.gob.es/ecieMaps/browser/metabuscador.html

Medical Concept Representation: Transform codes into numerical data

Transforming codes into numerical data

- Traditional approaches use **one-hot encoding**.

Vocabulary:

Care unit	Description
CCU	Coronary care unit
CSRU	Cardiac surgery recovery unit
MICU	Medical intensive care unit
SICU	Surgical intensive care unit
TSICU	Trauma/surgical intensive care unit

FIRST_CAREUNIT	FIRST_CAREUNIT_CCU	FIRST_CAREUNIT_CSRU	FIRST_CAREUNIT_MICU	FIRST_CAREUNIT_SICU	FIRST_CAREUNIT_TSICU
SICU	0	0	0	1	0
CCU	1	0	0	0	0
MICU	0	0	1	0	0
MICU	0	0	1	0	0
MICU	0	0	1	0	0

Nominal categorical data in MIMIC

Transformed into

Features with numerical data:
A vector of size = the vocabulary size (5 in the example), with 0's and one 1.

Why one-hot encoding fails

- Each code → vector of size |V|, where V=size of the dictionary.
- Example:
  - ICD_A → [1, 0, 0, 0, ...]
  - ICD_B → [0, 1, 0, 0, ...]
- Problems:
  - Very high dimensional (|V| > 70,000)
  - Sparse representations
  - No notion of similarity
- Example: "Diabetes" and "Hyperglycemia" are unrelated in this space ❌

Transforming codes into numerical data

- Alternative: Word Embeddings.
- Key idea: codes that appear in similar contexts should have similar representations.
- Example:
  - Hypertension ↔ Stroke ↔ Cardiovascular disease

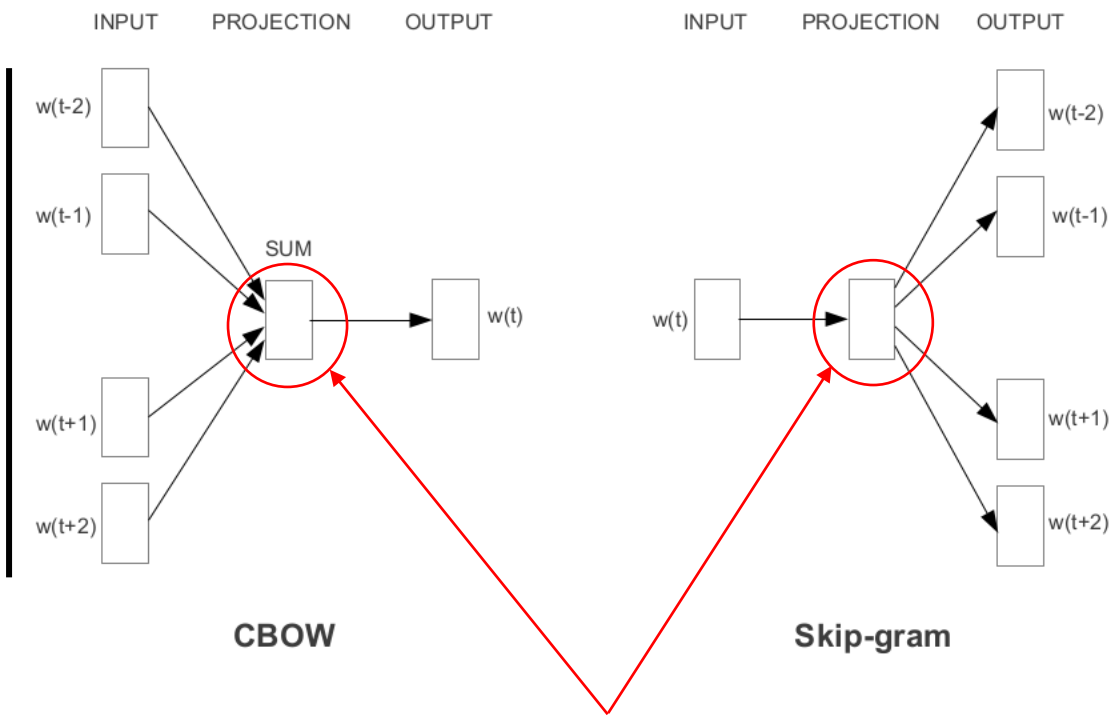
- Embeddings capture clinical relationships.

- Example:

  "Mr. Huntington was treated for Huntington's Disease at Huntington Hospital, located on Huntington Avenue."

Word Embeddings

- **Word2vec.** Two basic models for learning word embeddings:
  - **Skip-Gram**
  - Continuous Bag of Words (**CBOW**).
- A single-layer NN is trained to optimize the predictions, and its weight vector is taken to be the **embedding of the word**.
- Example:



"word embedding vectors".

…treated for Huntington's Disease at…

-2 -1 1 2

→ In Skip-gram, our input and target word pair would be:

- (Huntington, treated),
- (Huntington, for),
- (Huntington, Disease),
- (Huntington, at).

Word2Vec T. Mikolov (2013). «Efficient Estimation of Word Representations in Vector Space». arXiv:1301.3781.

Word Embeddings with DL

- We can model these architectures as DL classification models.
- Extensions of the basic models for learning word embeddings are:
  - **GloVe** (Global Vectors for Word Representation) use Word-Word co-occurrence matrix of the whole corpus.
  - **Elmo** (Embeddings from Language Model) is a Bi-LSTM Language model.

  Transformer-based architectures:
  - **BERT** (Bidirectional Encoder Representations from Transformers)
  - **GPT** (Generative Pre-trained Transformer)

Word Embeddings in EHR

- **Challenge**: Medical codes ≠ natural language
- No clear order of codes.

- Solutions:
  1. Adapt NLP models (Word2Vec, BERT)
     - Examples of biomedical specific word embeddings used to represent EHR terms trained on large PubMed datasets (http://bio.nlplab.org).

  2. Use medical structure
     - As method Med2Vec

Med2Vec (Choi et al.)

- **Key idea**:
  - Visits have temporal order → Learn representations at the VISIT level.
- Input:
  - Medical codes per visit + Patient demographics.
- Learn:
  - Code embeddings.
  - Visit embeddings.
- Use:
  - Predict next visit.
- **Advantage**:
  - Captures clinical progression over time.

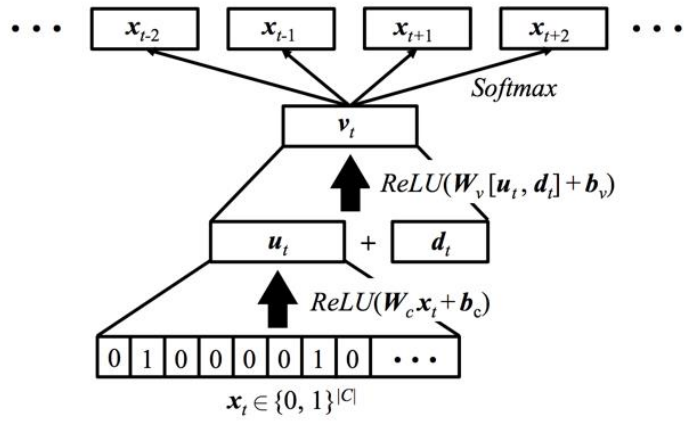


Figure 2: Structure of `Med2Vec`: A visit comprised of several medical codes is converted to a binary vector $\boldsymbol{x}_t \in \{0,1\}^{|C|}$. The binary vector is then converted to an intermediate visit representation $\boldsymbol{u}_t$. $\boldsymbol{u}_t$ is concatenated with a vector of demographic information $\boldsymbol{d}_t$, and converted to the final visit representation $\boldsymbol{v}_t$, which is trained to predict its neighboring visits $\ldots, \boldsymbol{x}_{t-2}, \boldsymbol{x}_{t-1}, \boldsymbol{x}_{t+1}, \boldsymbol{x}_{t+2}, \ldots$

Patient Representation

Patient State Representation

- Traditional approach are handcrafted features (in ICU):
  - APACHE (Acute Physiology and Chronic Health Evaluation) score
  - SOFA (Sequential Organ Failure Assessment) score
  - And others
- Pros:
  - Interpretable
  - Clinically validated
- Cons:
  - Limited scalability
  - Miss hidden patterns
- Motivation for representation learning

Patient Representation Learning

- Learning a **dense mathematical representation** of a patient that encodes meaningful information from EHR.

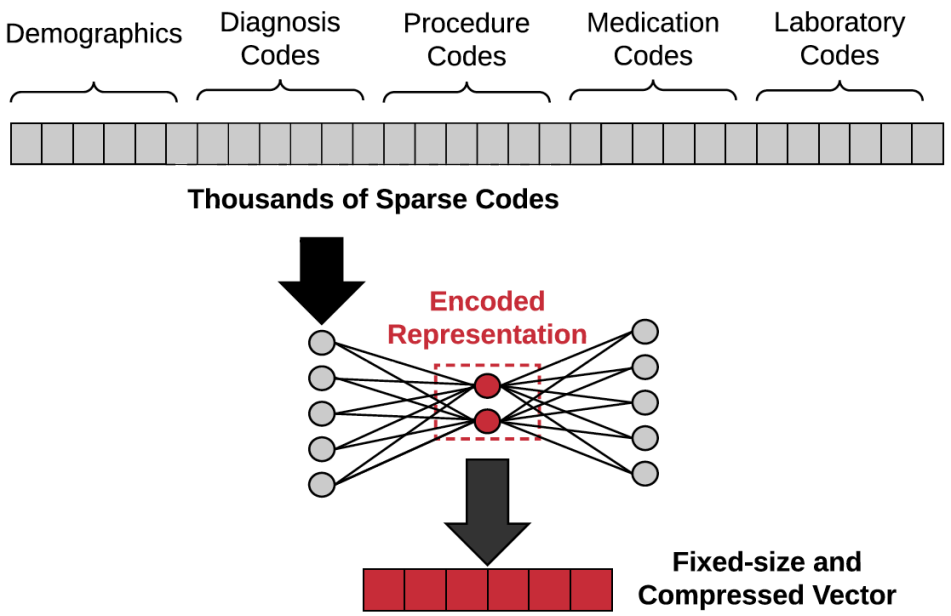
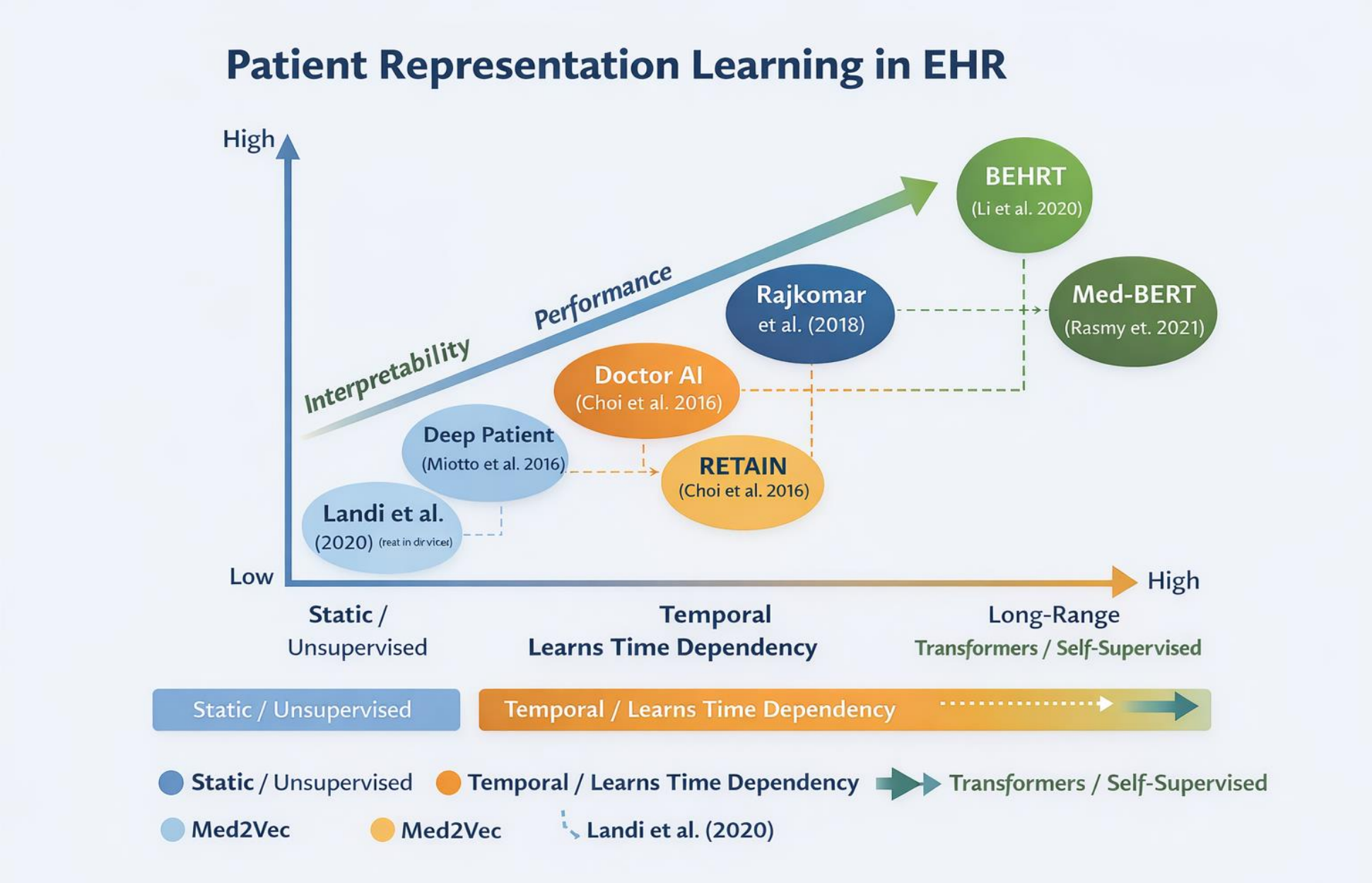


Illustration of autoencoders for patient representation.
Source: Schicle *et al.* arXiv:1706.03446v2

Patient Representation Learning: Evolution

1. Pre-2010:
   - Rule-based, expert features
2. 2010–2015:
   - Embeddings (Word2Vec, Med2Vec)
3. 2015–2020:
   - RNNs / LSTMs (Doctor AI)
4. 2020–present:
   - Transformers (BEHRT, ClinicalBERT)
   - Graph Neural Networks
   - Multimodal learning

Patient Representation Learning: Key Papers



Information Extraction in EHR

Information Extraction in HER

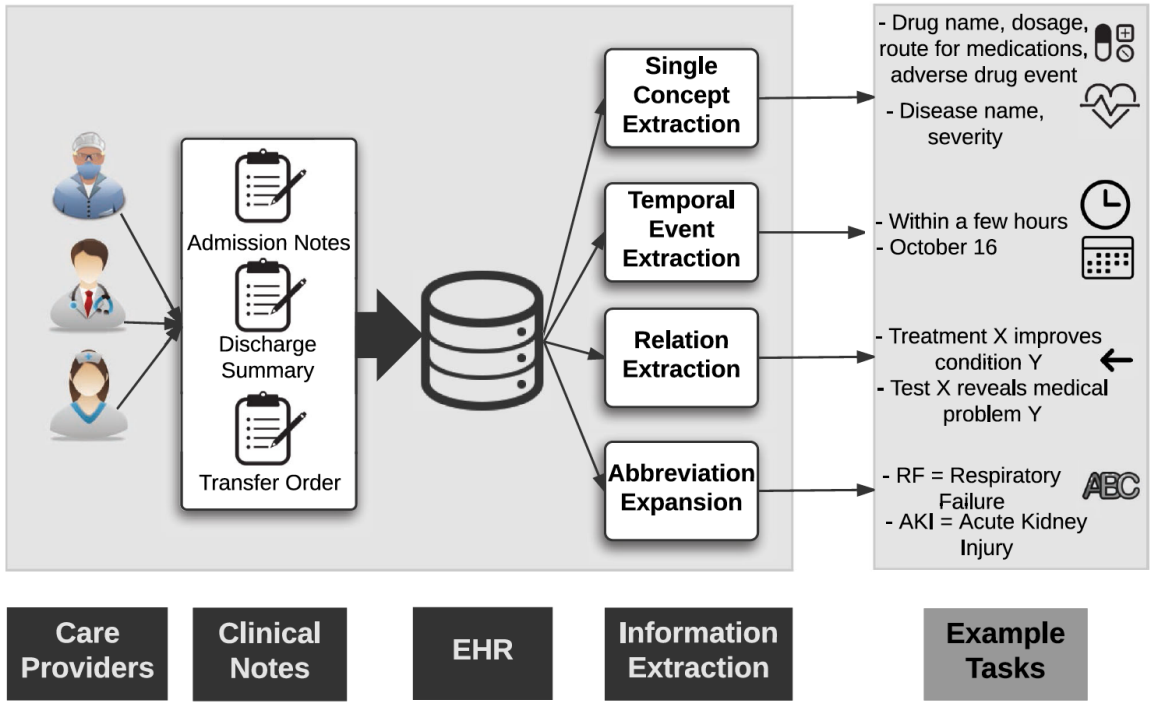
- Problem: Unstructured clinical text

- Examples:
  - Physician notes
  - Discharge summaries
  - Radiology reports

- Contains critical information NOT in structured fields

Information Extraction in EHR: Tasks

- Main NLP tasks in EHR:
  - Named Entity Recognition (NER)
  - Relation extraction
  - Clinical coding
  - Text similarity



EHR IE and example tasks. Source: Schicle *et al.* arXiv:1706.03446v2

Text representation

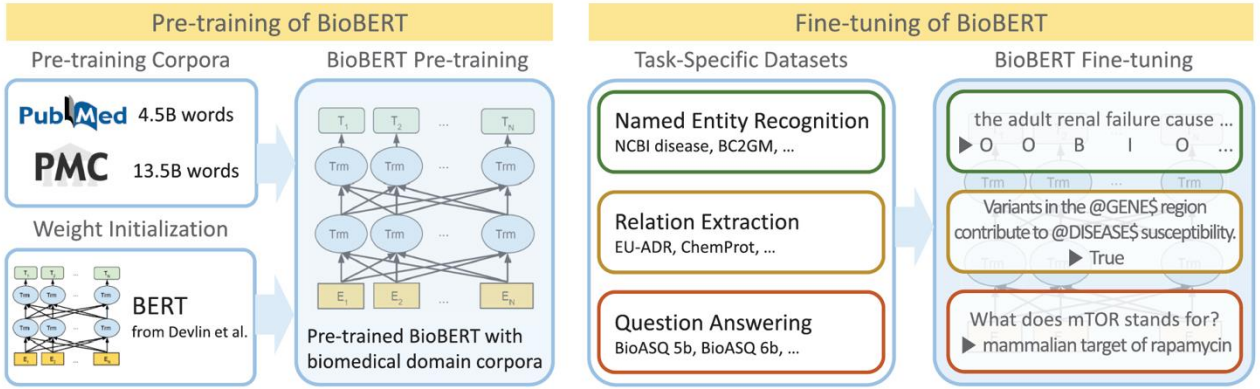
- Searching for similar or related clinical notes (patients).
- Detecting errors in the diagnosis report by looking for big changes in representations.

Information Extraction in EHR: Modern Approaches

Transformer-based models:

- BERT
- Clinical BERT embeddings (https://github.com/EmilyAlsentzer/clinicalBERT)
- BioBERT (https://github.com/dmis-lab/biobert), trained on a general domain corpus and fine-tuned on biomedical domain text.
- BioGPT

- Current standard in clinical NLP



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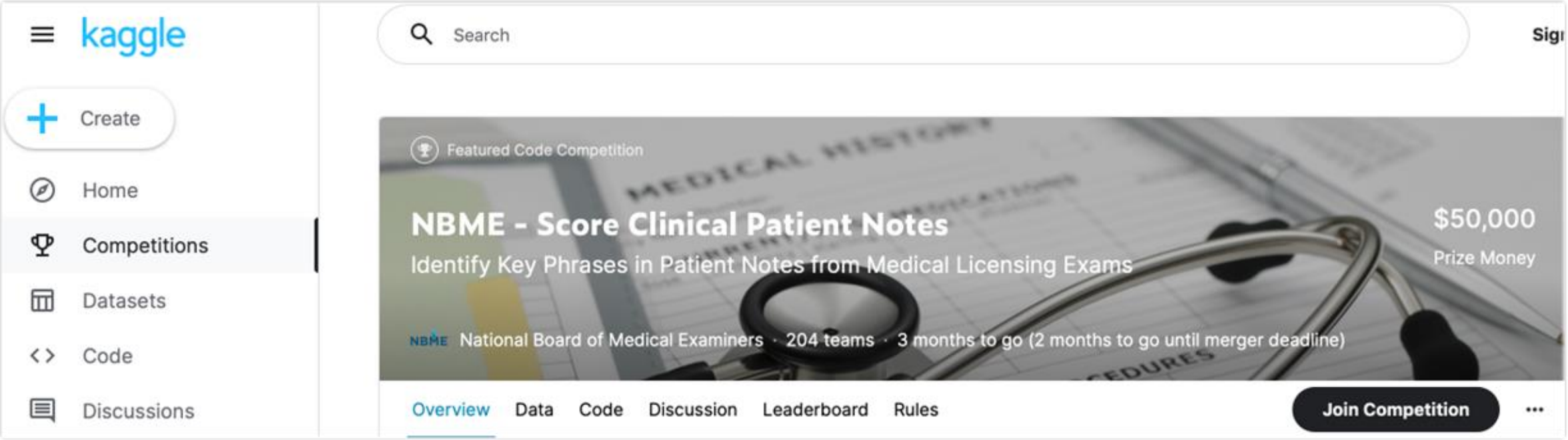
Challenges



- Competition to identify specific clinical concepts in patient notes.

Challenges

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